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COURSE: DSA-0504 QUERY PROCESSING FOR DATA SCIENCE

21. Implement pandas program to swap the cases of specified character

```
21-Q.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/21-Q.py (3.13.9)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    "Name": ["Alice", "Bob", "Charlie", "David"],
    "Gender": ["Female", "Male", "Female", "Male"]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

df["Gender"] = df["Gender"].str.swapcase()

print("\nAfter Swapping Cases:")
print(df)
```

OUTPUT:

```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/21-Q.py =====
Original DataFrame:
   Name  Gender
0  Alice  Female
1   Bob   Male
2 Charlie  Female
3  David   Male

After Swapping Cases:
   Name  Gender
0  Alice  fEMALE
1   Bob   mALE
2 Charlie  fEMALE
3  David   mALE
```

22. Draw a line with suitable cases of a specified character column

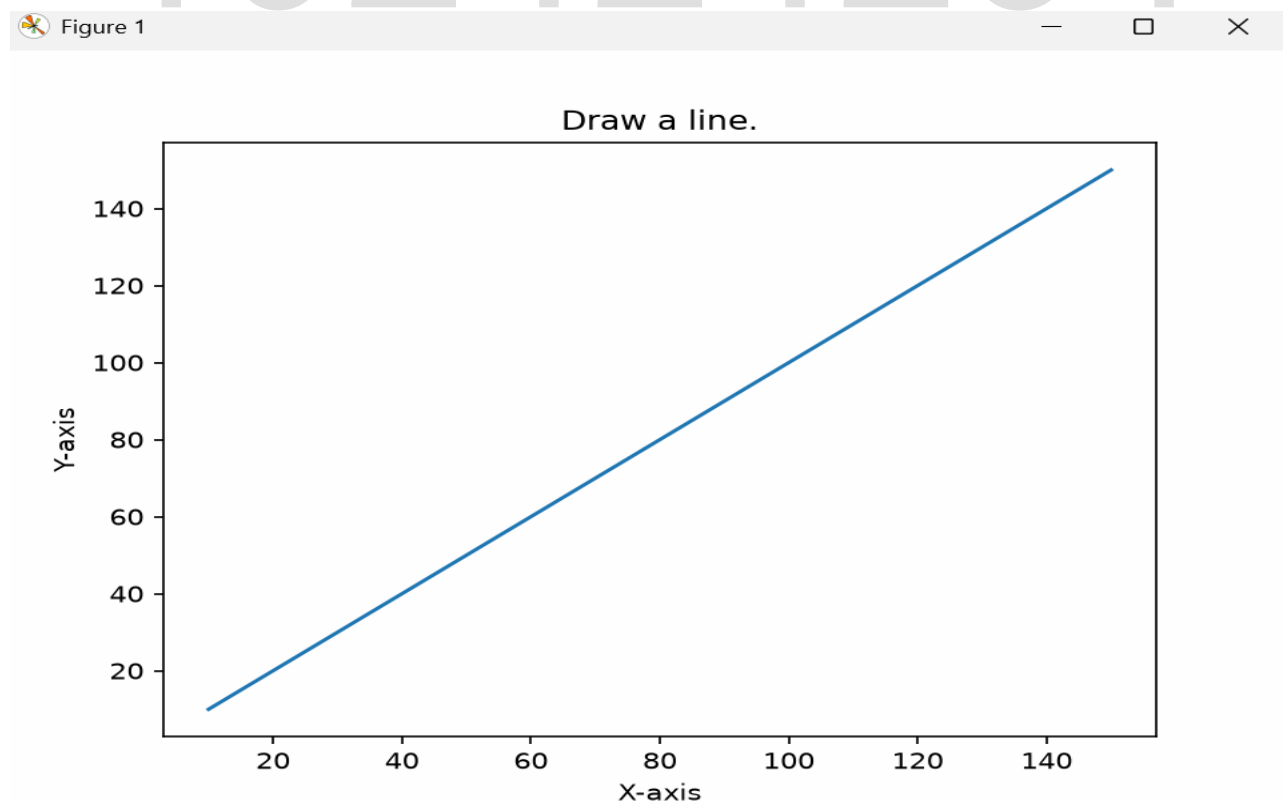
```
22-q.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/22-q.py (3.13.9)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

x = [10, 30, 50, 70, 90, 110, 130, 150]
y = [10, 30, 50, 70, 90, 110, 130, 150]

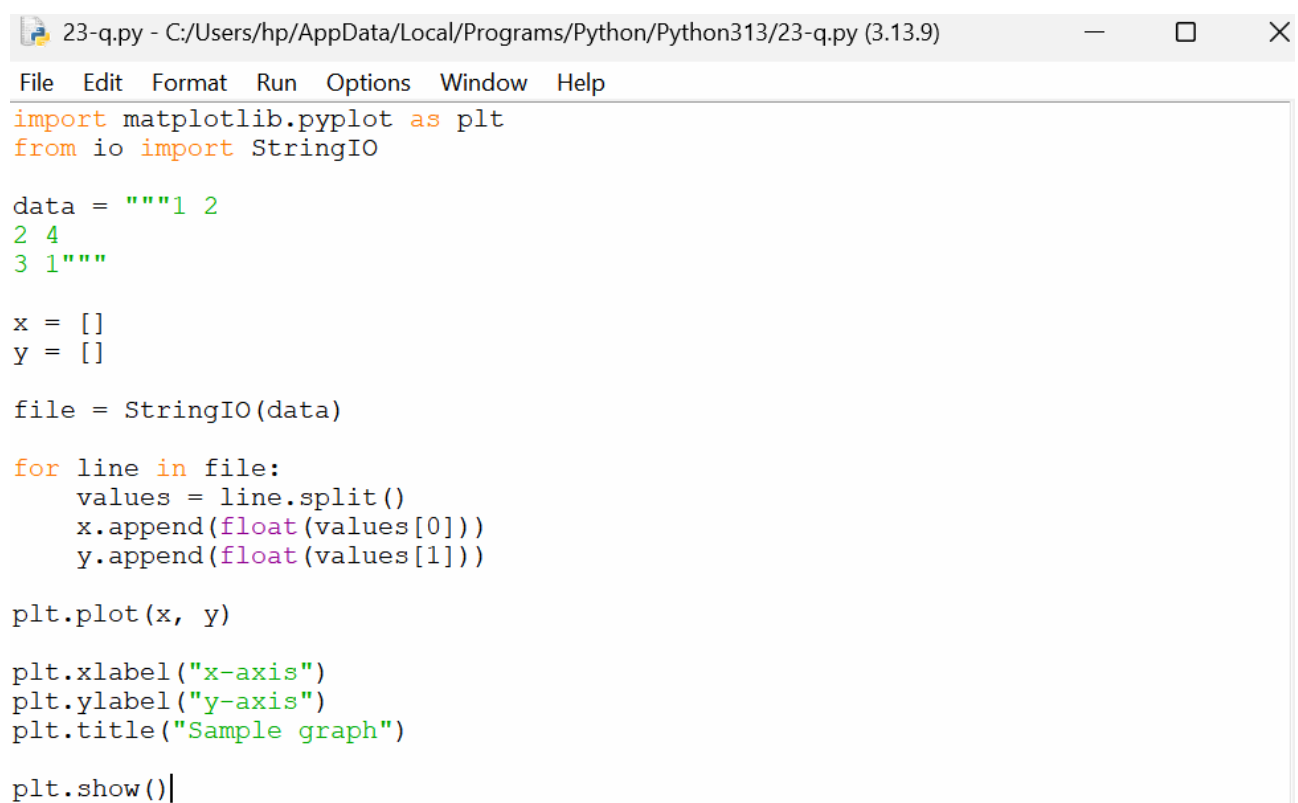
plt.plot(x, y)

plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Draw a line.")

plt.show()
```



23. Draw a line using given axis values



```
23-q.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/23-q.py (3.13.9)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from io import StringIO

data = """1 2
2 4
3 1"""

x = []
y = []

file = StringIO(data)

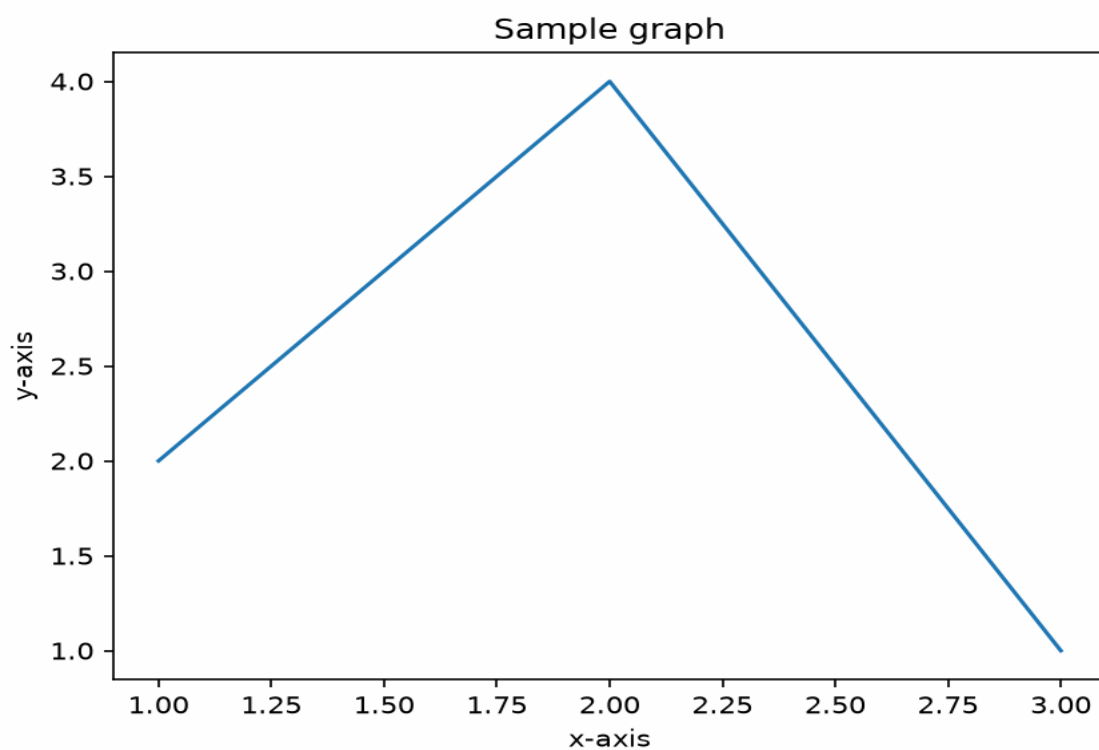
for line in file:
    values = line.split()
    x.append(float(values[0]))
    y.append(float(values[1]))

plt.plot(x, y)

plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Sample graph")

plt.show()
```

Figure 1



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